Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 29: Approved Solutions

Other Starting Points, Inequalities, and Divisibility

Covers Chartrand & Zhang, Section 4.2

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Three things that change from Lecture 28

1. **The base step may not be at 1.** Test small values first; start where the claim actually becomes true.
2. **Inequalities chain.** Start from one side of $P(k+1)$, bring in the hypothesis, and justify every link until you reach the other side.
3. **Divisibility needs the k case to appear.** Rewrite the $k+1$ expression so that the k expression shows up inside it — then substitute.

Three mistakes to expect

1. **Starting at $n=1$ for a claim false at $n=1$.**
2. **An unjustified link** in an inequality chain.
3. **Dividing by something that might be zero** or flipping an inequality by multiplying by a negative.

Part II: Practice Problems

Section A — Inequalities

1. Prove that $3^n \geq 2n+1$ for every positive integer n.

Approved solution

Proof. We proceed by induction. **Base step:** $3^1 = 3 \geq 3 = 2(1)+1$. ✓

Inductive step. Assume $3^k \geq 2k+1$ for a positive integer k. Then

$$3^{k+1} = 3 \cdot 3^k \;\geq\; 3(2k+1) = 6k+3 \;\geq\; 2k+3 = 2(k+1)+1.$$

The first inequality is the inductive hypothesis (multiplying by the positive number 3); the second holds because $6k \geq 2k$ for $k \geq 1$. By induction the result holds for all n. ■

Each link carries a reason. "$6k+3 \geq 2k+3$" is true but not obvious to every reader; saying why is part of the proof.

2. Test $n^2 > 2n+1$ for $n = 1, 2, 3, 4$. Identify where it starts to hold, and prove it from that point on.

Approved solution

Test: $n=1$: $1 > 3$ false. $n=2$: $4 > 5$ false. $n=3$: $9 > 7$ true. $n=4$: $16 > 9$ true. So the claim holds from $n = 3$.
Proof for $n \geq 3$. **Base step:** $9 > 7$. ✓
Inductive step. Assume $k^2 > 2k+1$ for an integer $k \geq 3$. Then

$$(k+1)^2 = k^2 + 2k + 1 \;>\; (2k+1) + 2k + 1 = 4k+2 \;\geq\; 2k+3 = 2(k+1)+1,$$

using the hypothesis for the first inequality and $2k \geq 1$ for the last. ■

The testing was not optional. Without it you might have tried a base step at $n = 1$, where the claim is false and the proof would have nothing to stand on.

Section B — Divisibility

3. Prove that 5 divides $6^n - 1$ for every positive integer n.

Approved solution

Proof. We proceed by induction. **Base step:** $6 - 1 = 5$. ✓
Inductive step. Assume $6^k - 1 = 5m$ for some integer m. Then

$$6^{k+1} - 1 = 6 \cdot 6^k - 1 = 6(6^k - 1) + 5 = 6(5m) + 5 = 5(6m+1).$$

Since $6m+1$ is an integer, $5 \mid 6^{k+1} - 1$. By induction the claim holds. ■

The same trick as Lecture 29. Write $6 \cdot 6^k - 1$ as $6(6^k - 1) + 5$ so the k case appears. The leftover constant is always one less than the base — here $6 - 1 = 5$.

4. Prove that 6 divides $n^3 - n$ for every positive integer n.

Approved solution

Proof. We proceed by induction. **Base step:** $1 - 1 = 0 = 6 \cdot 0$. ✓
Inductive step. Assume $k^3 - k = 6m$ for some integer m. Then

$$(k+1)^3 - (k+1) = k^3 + 3k^2 + 3k + 1 - k - 1 = (k^3 - k) + 3k(k+1).$$

The first term is $6m$. For the second, $k(k+1)$ is a product of consecutive integers, hence even, say $k(k+1) = 2j$; so $3k(k+1) = 6j$. Therefore $(k+1)^3 - (k+1) = 6m + 6j = 6(m+j)$. By induction the claim holds. ■

The inductive step needed an earlier result — that consecutive integers multiply to an even number, from Lecture 24. Induction proofs routinely lean on facts proved another way.

Section C — Where the base goes

5. A student proves $n! > 3^n$ "for every integer $n \geq 1$" by checking only the inductive step. Test the claim for small n, find where it actually begins, and identify exactly what the student's proof would have missed.

Approved solution

Test: $1! = 1 < 3$; $2! = 2 < 9$; $3! = 6 < 27$; $4! = 24 < 81$; $5! = 120 < 243$; $6! = 720 < 729$; $7! = 5040 > 2187$.
So $n! > 3^n$ first holds at $n = 7$, and is **false** for $n = 1, \ldots, 6$.
The inductive step does work for $k \geq 3$: $(k+1)! = (k+1)k! > (k+1)3^k \geq 3 \cdot 3^k$ once $k+1 \geq 3$. But the claim "for every $n \geq 1$" is false, and a proof with no base step cannot detect that.

The correct statement is "for every $n \geq 7$," with base step $7! > 3^7$. Six consecutive failures before the first success — the base step is not a formality, it is where the truth of the claim is anchored.

Section D — Writing

6. In five to eight complete sentences, explain how the proof that $2^n > n^2$ for $n \geq 5$ settles Practice Set 1, Problem 15. Say what you could and could not conclude in Week 1, and what induction adds.

Approved solution

Model answer. In Practice Set 1 we tested $2^n > n^2$ and found it true at $n = 1$, false at $n = 2, 3, 4$, and true at every value we tried from 5 upward. That was evidence, but it could not establish the claim for all $n \geq 5$, because there are infinitely many such integers and we had checked only finitely many. Induction supplies what was missing: the base step verifies $n = 5$, and the inductive step proves, for an arbitrary $k \geq 5$, that $2^k > k^2$ implies $2^{k+1} > (k+1)^2$. That second step is a general argument, not a check, so together with the base it covers every integer from 5 onward. The Week 1 behaviour also explains why the base step is 5: the claim is genuinely false at 2, 3, and 4, so no correct induction could begin earlier. What was a suspicion in Week 1 is now a theorem.

Full credit requires naming what Week 1 could not conclude and why, and identifying the inductive step as the general argument that closes the gap.

Instructor notes

- **Problem 6 is the loop-closer.** It asks students to connect Week 11 back to Week 1 explicitly. Worth reading a strong answer aloud.
- **Problem 5** shows six failures before the first success, and makes the case for testing before choosing a base step better than any lecture could.
- **Problem 4** leans on a Lecture 24 result inside an inductive step. Point it out: induction does not work in isolation.
- **Problem 2** requires finding the base step, not being given it.

- Reserved for Homework 11: Section 4.2 even-numbered exercises.